Influence of Atmospheric Turbulence Power Spectral Density Models on Orbital Angular Momentum Mode Coupling

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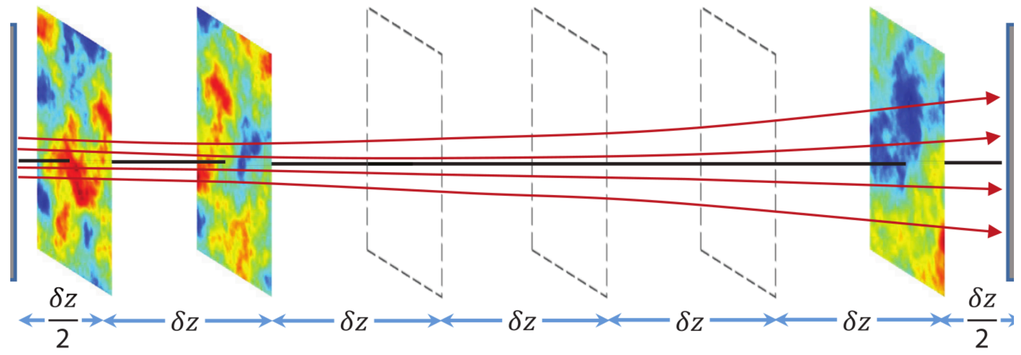
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At the same refractive-index structure constant C_n^2, the choice of turbulence PSD changes simulated OAM results by up to 25 %, and each spectral band affects a different metric.

Motivation

- OAM enables mode-division multiplexing in free-space optical links.
- Turbulence distorts the optical wavefront and causes modal crosstalk.
- **How does the turbulence PSD choice affect OAM metrics?**

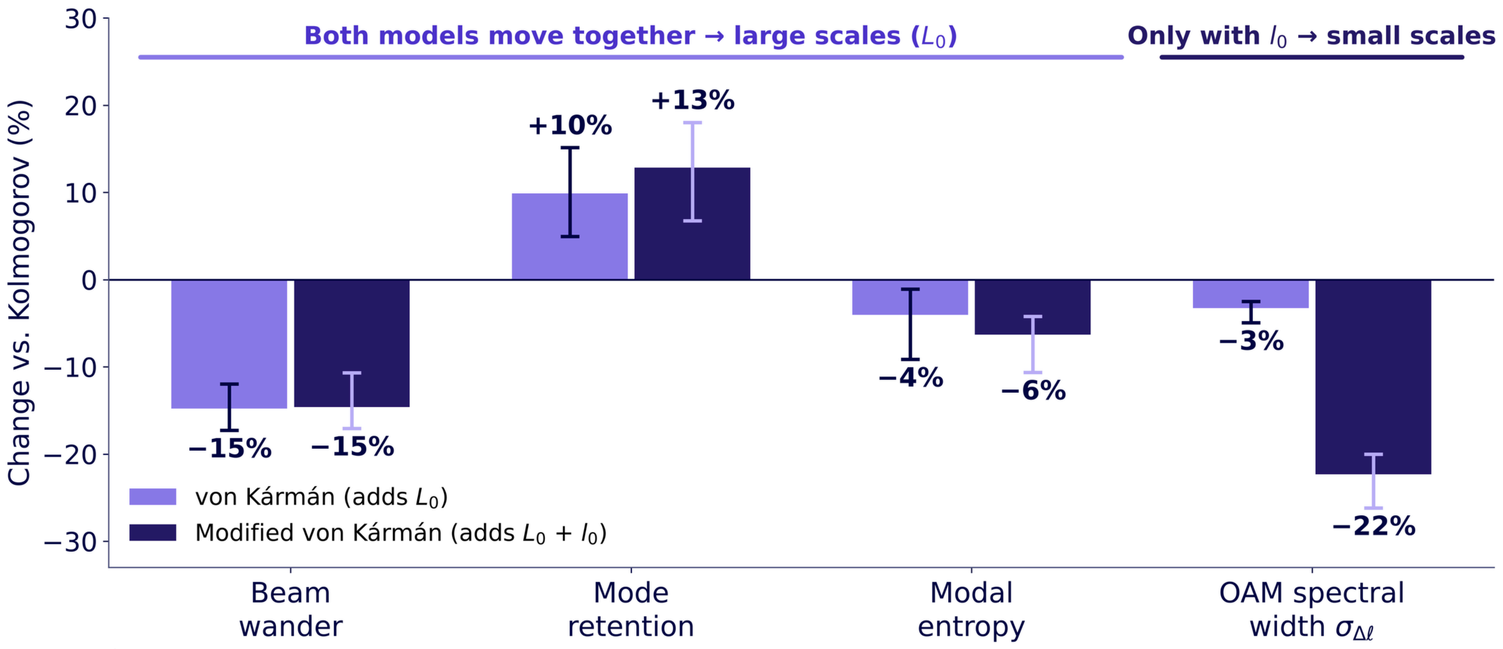
Method

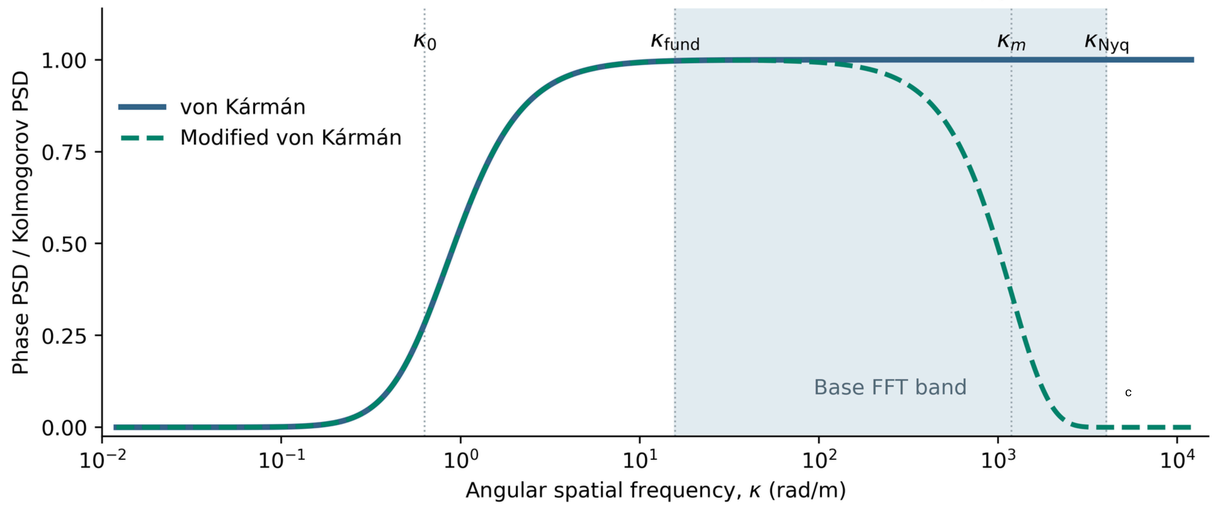


- On axis scintillation vs. Rytov theory **3,5%**
- RMS beam wander vs. analytic prediction **0,4%**
- OAM retention vs. Paterson model **3,4%**

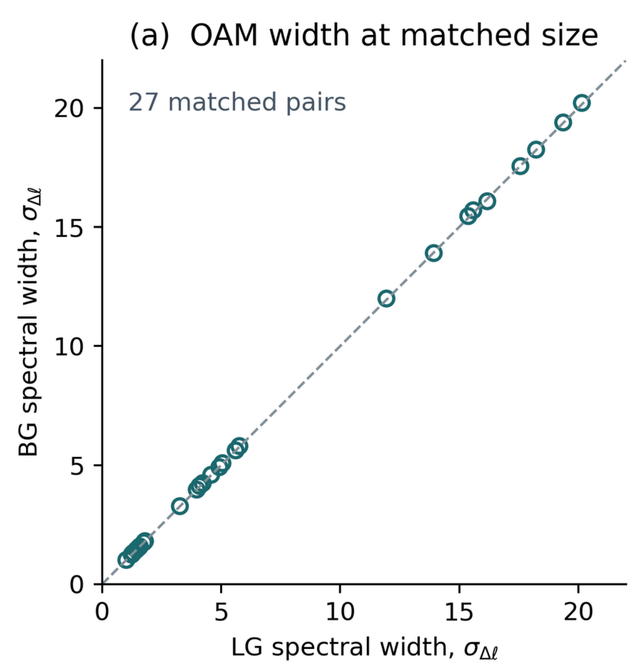
LG and BG beams, |ℓ| = 1,2, 3 (matched RMS size) → **Split-Step (ASM propagation + FFT phase screen)** → **OAM Spectrum P(ℓ) (Azimutal decomposition)**

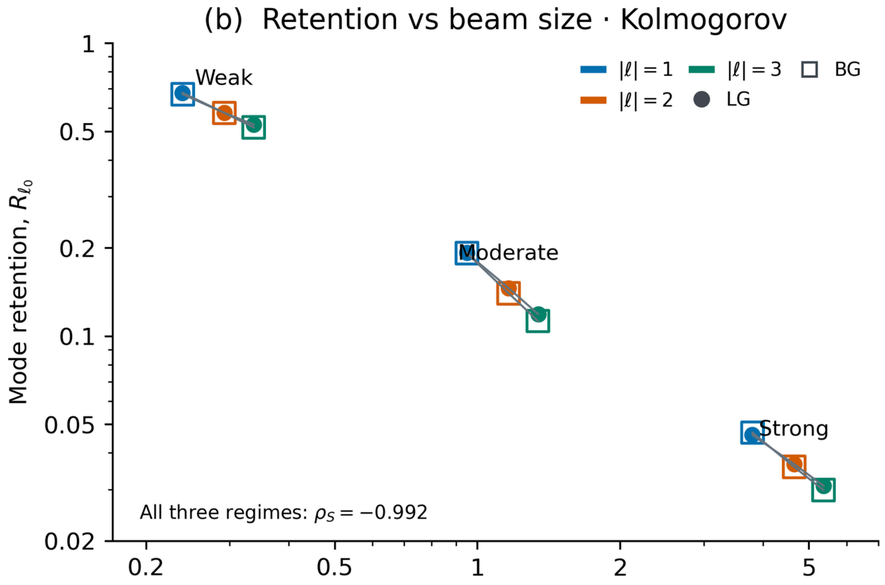
Large scales move the beam and small scales spread the OAM spectrum



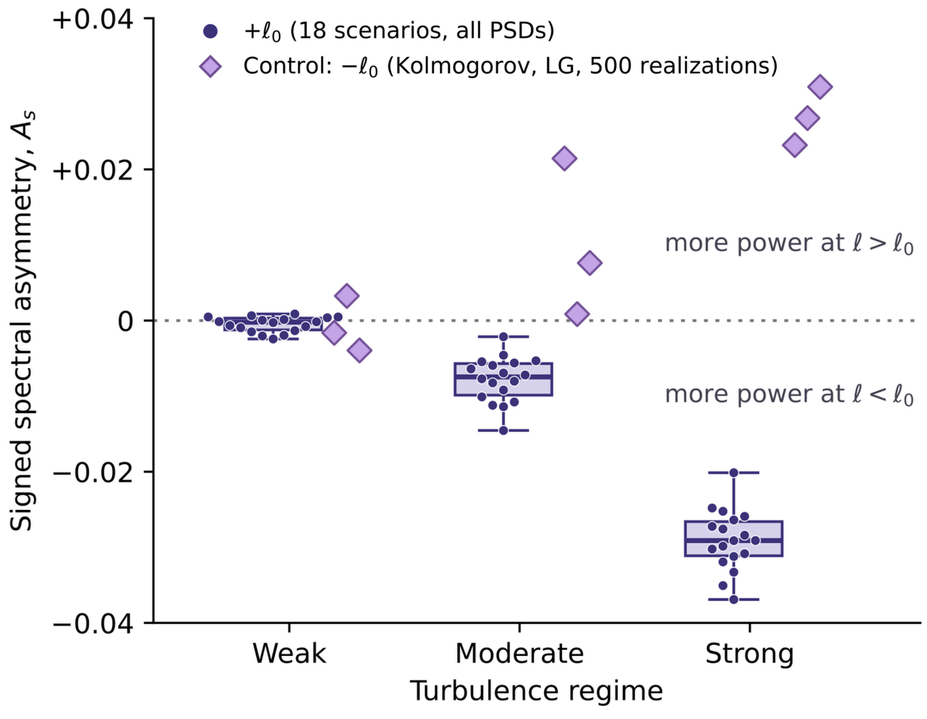


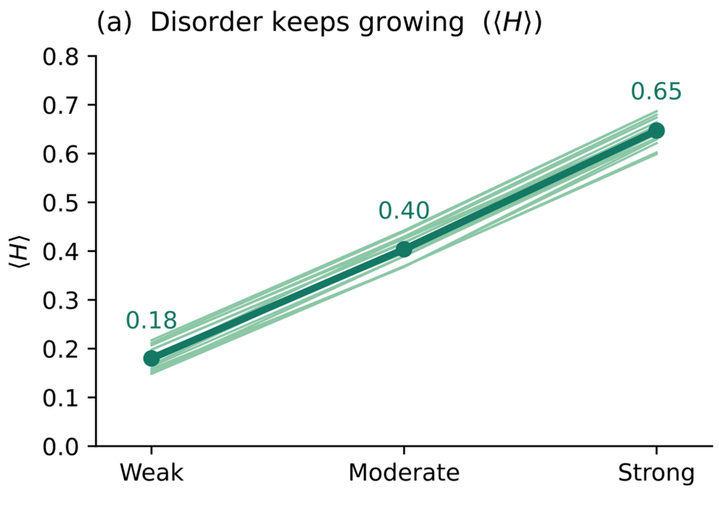
Beam size, not mode family

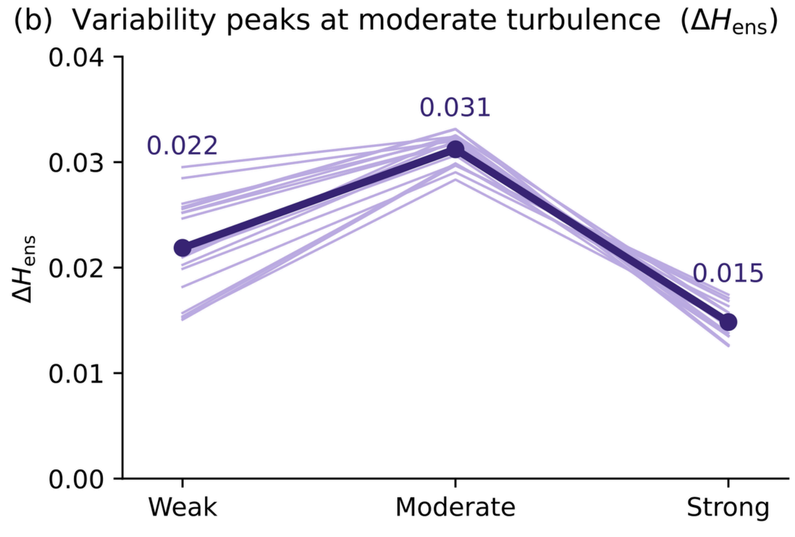




Further findings







[1] Andrews & Phillips, Laser Beam Propagation through Random Media, SPIE, 2005. – [2] Paterson, Phys. Rev. Lett., 94, 153901 (2005). – [3] Peters et al., Adv. Opt. Photonics, 17, 113–184 (2025). – [4] Klug et al., New J. Phys., 23, 093012 (2021). – [5] Ferlic et al., J. Opt. Soc. Am. A, 38(10), 1423–1437 (2021). – [6] Khare et al., Orbital Angular Momentum States of Light, IOP Publishing (2020).